Switch Questions

4 questions | Original English questions and source answers

Question 1

What are two functions of a Layer 2 switch? (Choose two)

Options

A. selects the best route between networks on a WAN

B. moves packets between different VLANs

C. acts as a central point for association and authentication servers

D. moves packets within a VLAN

E. makes forwarding decisions based on the MAC address of a packet

Answer

D. moves packets within a VLAN

E. makes forwarding decisions based on the MAC address of a packet

Source: CCNA 200-301 Questions.html | Question 1

Question 2

The SW1 interface g0/1 is in the down/down state. Which two configurations are valid reasons for the interface conditions? (Choose two)

Options

A. The interface is error-disabled

B. The interface is shut down

C. There is a protocol mismatch

D. There is a duplex mismatch

E. There is a speed mismatch

Answer

A. The interface is error-disabled

E. There is a speed mismatch

Source explanation

The down/down state indicates a Layer 1 problem -> “Protocol mismatch” answer is a Layer 2 problem so it is not correct.

Duplex should be matched on both side but it is not a must -> Therefore “duplex mismatch” is not a correct answer.

The most common Layer 1 problems are faulty or incorrect cabling or hardware failure but none of them is in the answer choices.

If the port is error-disabled, we will see it in down/down state so this answer is correct.

SW#sh int f1/0/1
FastEthernet1/0/1 is down, line protocol is down (err-disabled)

There are only two answers left, which are “speed mismatch” and “interface is shut down”. According to the table below (from the old and retired ICND1 100-101 Official Cert Guide), speed mismatch is also a cause of down/down state.

Table 10-4 LAN Switch Interface Status Codes
| Line Status | Protocol Status | Interface Status | Typical Root Cause

| Administratively Down | Down | disabled | The interface is configured with the shutdown command.

| Down | Down | notconnect | No cable; bad cable; wrong cable pinouts; the speeds are mismatched on the two connected devices; the device on the other end of the cable is (a) powered off, (b) shutdown, or (c) error disabled.

| Up | Down | notconnect | An interface up/down state is not expected on LAN switch physical interfaces.

| Down | Down (err-disabled) | err-disabled | Port security has disabled the interface.

| Up | Up | connected | The interface is working.

Note:
+ Speed must match between two ends. We tested speed mismatch on real device but we only received “up/down” state.
+ If “interface is shut down” then we will see the “administratively down/down”, not “down/down” state. But the “down/down” state would be seen on the far end interface.

The down/down state indicates a Layer 1 problem -> “Protocol mismatch” answer is a Layer 2 problem so it is not correct.

Duplex should be matched on both side but it is not a must -> Therefore “duplex mismatch” is not a correct answer.

The most common Layer 1 problems are faulty or incorrect cabling or hardware failure but none of them is in the answer choices.

If the port is error-disabled, we will see it in down/down state so this answer is correct.

```
SW#sh int f1/0/1
FastEthernet1/0/1 is down, line protocol is down (err-disabled)
```

There are only two answers left, which are “speed mismatch” and “interface is shut down”. According to the table below (from the old and retired ICND1 100-101 Official Cert Guide), speed mismatch is also a cause of down/down state.

Table 10-4 LAN Switch Interface Status Codes

Line Status	Protocol Status	Interface Status	Typical Root Cause
Administratively Down	Down	disabled	The interface is configured with the **shutdown** command.
Down	Down	notconnect	No cable; bad cable; wrong cable pinouts; the speeds are mismatched on the two connected devices; the device on the other end of the cable is (a) powered off, (b) **shutdown**, or (c) error disabled.
Up	Down	notconnect	An interface up/down state is not expected on LAN switch physical interfaces.
Down	Down (err-disabled)	err-disabled	Port security has disabled the interface.
Up	Up	connected	The interface is working.

Note:
+ Speed must match between two ends. We tested speed mismatch on real device but we only received “up/down” state.
+ If “interface is shut down” then we will see the “administratively down/down”, not “down/down” state. But the “down/down” state would be seen on the far end interface.

Source: CCNA 200-301 Questions.html | Question 2

Question 3

How does the dynamically-learned MAC address feature function?

Options

A. It requires a minimum number of secure MAC addresses to be filled dynamically

B. The ports are restricted and learn up to a maximum of 10 dynamically-learned addresses

C. Switches dynamically learn MAC addresses of each connecting CAM table

D. The CAM table is empty until ingress traffic arrives at each port

Answer

D. The CAM table is empty until ingress traffic arrives at each port

Source explanation

The MAC addresses in the CAM table are the source MAC addresses only. Therefore it only learns MAC address from ingress traffic.

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Which three statements about MAC addresses are correct? (Choose three)

Options

A. The MAC address of a device must be configured in the Cisco IOS CLI by a user with administrative privileges

B. A MAC address contains two main components, the first of which identifies the network on which the host resides and the second of which uniquely identifies the host on the network

C. A MAC address contains two main components, the first of which identifies the manufacturer of the hardware and the second of which uniquely identifies the hardware

D. An example of a MAC address is 0A:26:B8:D6:65:90

E. To communicate with other devices on a network, a network device must have a unique MAC address

F. The MAC address is also referred to as the IP address

Answer

C. A MAC address contains two main components, the first of which identifies the manufacturer of the hardware and the second of which uniquely identifies the hardware

D. An example of a MAC address is 0A:26:B8:D6:65:90

E. To communicate with other devices on a network, a network device must have a unique MAC address

Source: CCNA 200-301 Questions.html | Question 4